Exam Date & Time: 26-Dec-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH BIOMEDICAL ENGINEERING END SEMESTER MAKE-UP EXAMINATIONS, DEC 2024

ELECTRONICS CIRCUITS [BME 2122]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

Missing data may be suitably assumed
Draw neat circuit diagrams wherever necessary

1) Build a tuned oscillator in which reactance X_1 & X_2 are inductive and the reactance X_3 is capacitive. Explain its working. (4)

A)

B) Compare four types of negative feedback topologies with appropriate block diagrams and characteristics. (4)

C) Making use of an operational amplifier build a Schmitt trigger circuit. (2)

2) How do you build monostable multivibrator using an operational amplifier? Explain in detail. (5)

A)

B) Design a square wave oscillator for a frequency f_o = 1 kHz & V_o = ±6.7V using the 741 op-amp and DC supply of voltage ± 12 V. (3)

C) Making use of an op-amp zero cross detector build a timing marker signal generator. (2)

3) Is it possible to use an operational amplifier for (i) computing area under a curve or signal (ii) identification of sudden changes in the signal? Explain in detail. (5)

A)

B) Design a 555-timer based astable multivibrator to generate a square of frequency 2 KHz and 60% duty cycle. Assume a capacitor of value 0.1 µF for the timing network. (3)

C) How do you build a pulse stretcher using 555 timer? Explain. (2)

4) Construct a positive edge triggered 555 timer monostable multivibrator and explain its operation. (3)

A)

B) How do you obtain load regulation characteristics of voltage regulator IC the 7905? (3)

Explain.

C) Design a 500mA constant current source using voltage regulator IC the 7805. Assume a quiescent current of 4.3mA. (4)

5) Construct an 8-bit binary weighted DAC using an Op-amp and explain its operation. (4)

A)

B) Build a counter type ADC and explain its operation. (4)

C) Compute the resolution & Full-scale output of an 8-bit R-2R ladder DAC. Assume V_R=5V and R_F=R. (2)

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